

Kyaw Min Thu

Automation Controls Engineer

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SUMMARY

Controls engineer with 20+ years across offshore drilling, semiconductor fabs, EV/solar manufacturing, high-pressure R&D;, and large-scale fulfillment automation. Fluent in Rockwell, Siemens, and Mitsubishi PLC platforms; bridges hardware-level commissioning with Python-based data tooling.

TECHNICAL SKILLS

PLC / Control Platforms	Rockwell ControlLogix / CompactLogix, Siemens S7-300/1500, Mitsubishi FX, Omron
Programming	Ladder Logic, Structured Text, FBD · Python, SQL, C · LabVIEW
Systems & Protocols	VFDs, Servo Drives, Fanuc Robotics, NI CompactDAQ · EtherNet/IP, Modbus TCP, OPC UA, MQTT
Other	Electrical schematics, panel fabrication, safety interlocks, ATEX, SCADA/HMI, data acquisition

EXPERIENCE

Automation Engineer — Controls Lead C&W; Services / JLL · Amazon Fulfillment Centers - Controls lead for Rockwell ControlLogix PLCs on conveyor and sortation systems across Amazon FC sites. - Primary technical lead for Fanuc robot troubleshooting, commissioning, and program modifications. - Built Python fault-trending tools to identify downtime patterns and reduce repeat failures. - Commissioned conveyor inspection system integrating sensors, PLC logic, and HMI diagnostics.	Mar 2025 – Present
Electrical Technician — Automation & Controls Energy Recovery Inc. - Designed NI CompactDAQ / LabVIEW control systems for R&D; high-pressure pump and valve test rigs. - Programmed Allen-Bradley PLCs (ControlLogix, CompactLogix); integrated Keyence vision into test workflows. - Full lifecycle ownership: schematics, BOM, panel build, safety interlocks, installation, commissioning. - Python + SQL data acquisition pipelines for experiment traceability and regulatory reporting.	Mar 2018 – Feb 2025
Maintenance Technician — Automation & Controls Tesla Motors / SolarCity - Supported high-volume EV and solar manufacturing automation (Fanuc robotics, Rockwell PLCs). - Data-driven MTBF improvement programs; modified control logic for production startups and line expansions.	Oct 2015 – Mar 2018
Field Service Engineer — Controls & Automation MHWirth (Singapore) Pte. Ltd. - Led full PLC conversion of offshore drilling top drive (Mitsubishi → Siemens S7): I/O mapping, interlock logic, ATEX cabinet fabrication, and offshore commissioning. - Tuned hydraulic PID loops and verified performance under live load conditions on active rigs.	Apr 2012 – Apr 2015
System Engineer Hexcel Solutions Pte. Ltd. Designed and commissioned automated firefighter training systems for Singapore Civil Defence Academy; PLC programming and SCADA for fire, gas, and hydraulic control.	May 2009 – Oct 2010
Associate Engineer — Computer Integrated Manufacturing Chartered Semiconductor Manufacturing Maintained CIM systems (IBM SiView) and factory automation data platforms in a 300mm wafer fab.	Jun 2005 – Dec 2008

EDUCATION

BS Computer Science	California State University, East Bay
Diploma — Electronics, Computers & Communications Engineering	Singapore Polytechnic · Apr 2004